

equational logic, but any equational proof in $e_g(e)A$ can be turned into an equational proof in A after we replace the multiplicative identity with $e_g(a)$. Alternatively, we may argue as in the proof of Theorem 7.4 and assume $1 = e_{T^2+T+1}(a) = a^2 + a$. So we will assume

$$a^2 + a + 1 = 0$$

from now on. We enumerate the polynomials f_0, f_1, f_2, f_3 over $\mathbb{F}_2$ of degree < 2 by $0, 1, T, T+1$. If $b \in A$, we define a sequence recursively by $b_0 := b$ and $b_{n+1} := ab_n + b_na + f_n(a)b_n$. We have $b_0 = b$, $b_1 = ab + ba$ and then

$$\begin{aligned} b_2 &= ab_1 + b_1a + b_1 \\ &= a(ab + ba) + (ab + ba)a + (ab + ba) \\ &= a^2b + ba^2 + ab + ba \quad (\text{using } a^2 = a + 1) \\ &= (a + 1)b + b(a + 1) + ab + ba = 0. \end{aligned}$$

So we are lucky, the sequence terminates quite early. (We only expected $b_4 = 0$ from the general proof in Theorem 7.5.) Assume that we can show

$$ax + xa = x \implies x = 0.$$

Then we can derive $b_1 = 0$ from $b_2 = 0$ and we are done. This is exactly the constructive Wedderburn Theorem $W_{2,2,T+1}$. Let us prove it here: Assume $ax + xa = x$. We calculate

$$(a + x)^2 = a^2 + ax + xa + x^2 = a^2 + (x + x^2).$$

Since $x + x^2$ is idempotent, it is central. Hence, we can calculate

$$a + x = (a + x)^4 = (a^2 + (x + x^2))^2 = a^4 + (x + x^2)^2 = a + x + x^2.$$

From this we conclude $x^2 = 0$, i.e. $x = 0$, and we are done.

One could remove all references to the more general method from this proof, and also get rid of all remarks outside of equational logic. We will not do that here, also because it will be less clear what is actually happening (and why).

Example 7.8. For $p = 2$ and $k = 3$, one can proceed in a similar way, and we only briefly indicate how it works (we give a better proof in a moment). It suffices to consider $g = T^3 + T + 1$, since the other irreducible polynomial of degree 3 is $g(T + 1)$. So we may assume $a^3 + a + 1 = 0$. We enumerate the polynomials f_i of degree < 3 by $0, 1, T, T + 1, T^2, T^2 + 1, T^2 + T, T^2 + T + 1$. Define the sequence b_n recursively by $b_0 := b$ and $b_{n+1} := ab_n + b_na + f_n(a)b_n$. Then a routine calculation gives $b_7 = 0$ (even though we only expected $b_8 = 0$). Hence, if $W_{2,3,f}$ has been verified for all f of degree < 3 , we get an equational proof that 8-rings are commutative.

In many examples, one already has $b_n = 0$ for some $n < p^m$. This means that the commutativity proof does not require all $W_{p,k,f}$. Choosing a different enumeration of the f_i will also help to reduce the number n with $b_n = 0$. For example, in Example 7.8 we will get $b_3 = 0$ with a different enumeration. Actually, certain monomials are always sufficient:

Theorem 7.9. *Let $p \in \mathbb{P}$ and $k \geq 1$. Assume that for all $1 \leq i < k$ there is an equational proof of the constructive Wedderburn Theorem $W_{p,k,T^{p^i}}$. Then there is an equational proof that every p^k -ring with $p = 0$ is commutative.*

Proof. Let A be a p^k -ring with $p = 0$, and let $a, b \in A$. We will tweak the proof of Theorem 7.5. Let us first simplify the definition of b_n a bit. The old definition was $b_{n+1} = (a - f_n(a))b_n - b_na$. So we might as well define

$$b_0 := b, \quad b_{n+1} := b_na - f_n(a)b_n, \tag{17}$$

and start the enumeration with $f_0 := T$. Again, W_{p,k,f_n} proves $b_{n+1} = 0 \implies b_n = 0$ for $n \geq 1$ (using that $f_n(a) - a$ is a unit then), and we have $b_1 = ba - ab$. Hence, our goal is to

find an equational proof of $b_n = 0$ for some $n \geq 1$. As before, we may assume $g(a) = 0$ for some monic irreducible polynomial $g \in \mathbb{F}_p[T]$ whose degree m divides k .

Now, even though the computation of b_n takes place inside a ring that is non-commutative a priori, we shall see now that it can be encoded by a computation in a commutative ring. By induction, it follows from eq. (17) that we can express b_n as

$$b_n = \sum_{i,j} \lambda_{i,j} a^i b a^j$$

with $\lambda_{i,j} \in \mathbb{F}_p$. We encode this with the commutative polynomial

$$B_n := \sum_{i,j} \lambda_{i,j} X^i Y^j \in \mathbb{F}_p[X, Y].$$

More formally, we define $B_n \in \mathbb{F}_p[X, Y]$ recursively by

$$B_0 := 1, \quad B_{n+1} := B_n Y - f_n(X) B_n.$$

Then b_n results from B_n by replacing any monomial $\lambda X^i Y^j$ with $\lambda a^i b a^j$. This describes a map $\mathbb{F}_p[X, Y] \mapsto A$ that is clearly additive (but not multiplicative). In particular, $B_n = 0$ will imply $b_n = 0$. Since $\mathbb{F}_p[X, Y]$ is commutative, there is a very simple formula for B_n : With $B_0 = 1$ and $B_{n+1} = (Y - f_n(X)) B_n$ we find

$$B_n = \prod_{i < n} (Y - f_i(X)). \quad (18)$$

Consider the finite field $K := \mathbb{F}_p[X] / \langle g(X) \rangle$. (This concept is not equational, but wait for it.) The irreducible polynomial $g(Y) \in \mathbb{F}_p[Y] \subseteq K[Y]$ has a root in K , namely $[X]$. Basic finite field theory tells us that the roots are formed by the orbits under the Frobenius. Thus, we have $g(Y) = \prod_{i < m} (Y - [X]^{p^i})$ in $K[Y]$. This means that in $\mathbb{F}_p[X, Y]$ there is an equation of the form

$$g(Y) = \prod_{i < m} (Y - X^{p^i}) + g(X) \cdot h \quad (19)$$

for some $h \in \mathbb{F}_p[X, Y]$. For every specific choice of p, g , we can simply find this equation by calculating $\prod_{i < m} (Y - X^{p^i})$, so that no field theory is required. Define $f_i := T^{p^i}$ for $i < m$; the rest of the polynomials do not matter anymore. Then eqs. (18) and (19) imply

$$g(Y) = B_m(Y) + g(X) \cdot h.$$

But since $g(a) = 0$, we then get $b_m = 0$. For a purely equational proof, when p, g are fixed, one can just compute $b_m = 0$ directly, just by using $g(a) = 0$ (see the examples below). $\square$

Example 7.10. From Theorem 7.9 we get the following equational proof that 8-rings are commutative, for which only Morita [Mo78] gave an equational proof so far. We may assume $g(a) = 0$ for $g := T^3 + T + 1$. For any element b , we define a sequence recursively by $b_0 := b$ and $b_{n+1} := b_n a + a^{2^n} b_n$. We calculate:

$$\begin{aligned} b_1 &= ba + ab, \\ b_2 &= (ba + ab)a + a^2(ba + ab) \\ &= ba^2 + aba + a^2ba + a^3b \quad (\text{use } a^3 = a + 1) \\ &= b + ab + ba^2 + aba + a^2ba, \\ b_3 &= (b + ab + ba^2 + aba + a^2ba)a + (a + a^2)(b + ab + ba^2 + aba + a^2ba) \\ &= ba + aba + ba^3 + aba^2 + a^2ba^2 + ab + a^2b + aba^2 + a^2ba + a^3ba \\ &\quad + a^2b + a^3b + a^2ba^2 + a^3ba + a^4ba \\ &= ba + ab + aba + ba^3 + a^2ba + a^3b + a^4ba \quad (\text{use } a^3 = 1 + a, a^4 = a + a^2) \\ &= ba + ab + aba + b + ba + a^2ba + b + ab + aba + a^2ba \\ &= 0. \end{aligned}$$

If $W_{2,3,T^4}$ holds, $b_3 = 0$ implies $a^2b_2 = 0$, hence $b_2 = 0$. If $W_{2,3,T^2}$ holds, then $b_2 = 0$ implies $(a^2 + a)b_1 = 0$, hence $b_1 = 0$ and a, b commute. We will prove these constructive Wedderburn Theorems in the next section as part of more general results.

We can also replace the proof of $b_3 = 0$ with a “commutative computation”, as explained in the general proof in Theorem 7.9. Define polynomials $B_n \in \mathbb{F}_2[X, Y]$ recursively by $B_0 := 1$ and $B_{n+1} := B_n Y + X^{2^n} B_n$, so that b_n results from B_n by replacing any monomial $X^i Y^j$ with $a^i b a^j$. We compute

$$\begin{aligned} B_3 &= (Y + X)(Y + X^2)(Y + X^4) \\ &= Y^3 + (X + X^2 + X^4)Y^2 + (X^3 + X^4 + X^6)Y + X^7 \\ &= Y^3 + Xg(X)Y^2 + X^3g(X)Y + (X^4 + X^2 + X + 1)g(X) + 1 \\ &\equiv Y^3 + Y + 1 \pmod{g(X)} \\ &\equiv 0 \pmod{g(X), g(Y)} \end{aligned}$$

and hence $b_3 = 0$.

Example 7.11. With Theorems 7.5 and 7.9 we get the following equational proof that every 27-ring with $3 = 0$ is commutative. (There seems to be no equational proof in the literature yet.) The list of irreducible polynomials over $\mathbb{F}_3$ whose degree divides 3 is

$$\begin{aligned} &T, T + 1, T + 2, T^3 + 2T + 1, T^3 + 2T + 2, T^3 + T^2 + 2, T^3 + T^2 + T + 2, \\ &T^3 + T^2 + 2T + 1, T^3 + 2T^2 + 1, T^3 + 2T^2 + T + 1, T^3 + 2T^2 + 2T + 2. \end{aligned}$$

We may remove the linear polynomials and those polynomials which are linear substitutions of others. This leaves us with $T^3 + 2T + 1$ and $T^3 + T^2 + 2$. So we need to handle two cases.

Assume first that $a^3 + 2a + 1 = 0$. Let us choose an enumeration starting with $T, T + 1, T + 2$. We calculate:

$$\begin{aligned} b_1 &= ba - ab, \\ b_2 &= (ba - ab)a - (a + 1)(ba - ab) \\ &= ba^2 + aba + a^2b - ba + ab, \\ b_3 &= (ba^2 + aba + a^2b - ba + ab)a - (a + 2)(ba^2 + aba + a^2b - ba + ab) \\ &= ba^3 + aba^2 + a^2ba - ba^2 + aba - aba^2 - a^2ba - a^3b + aba - a^2b \\ &\quad + ba^2 + aba + a^2b - ba + ab \\ &= ba^3 - a^3b - ba + ab \quad (\text{use } a^3 = a - 1) \\ &= b(a - 1) - (a - 1)b - ba + ab = 0. \end{aligned}$$

Thus, if $W_{3,3,T+1}$ and $W_{3,3,T+2}$ hold, we get $b_1 = 0$, and we are done.

Now assume $a^3 + a^2 + 2 = 0$. Let us choose an enumeration starting with T, T^3, T^9 . We calculate (we omit the details):

$$\begin{aligned} b_1 &= ba - ab, \\ b_2 &= b_2a - a^3b_2 = b_2a - (1 - a^2)b_2 \\ &= -b + ab - ba + a^2b - aba + ba^2 + a^2ba, \\ b_3 &= b_3a - a^9b_3 = b_3a - (1 - a + a^2)b_3 = 0. \end{aligned}$$

Thus, if $W_{3,3,T^3}$ and $W_{3,3,T^9}$ hold, we get $b_1 = 0$, and we are done. We will prove the required Wedderburn Theorems in the next section.

Example 7.12. Let us prove that 16-rings are commutative with Theorem 7.9. (Again, there seems to be no equational proof in the literature yet.) The irreducible polynomials over $\mathbb{F}_2$